

LAB TASK # 8

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.MODEL SMALL

.STACK 100H

.DATA

values dw 100, 200, 300, 400

name_str db 'CLAUDE', 0

name_len equ \$ - name_str - 1

val1 dw 1234h

val2 dw 5678h

.CODE

MAIN PROC

MOV AX,@DATA

MOV DS,AX

CALL PROC1

call proc2

call proc3

MOV AH,4CH

INT 21H

MAIN ENDP

;END MAIN

PROC1 PROC

mov ax, values

push ax

mov ax, values+2

push ax

mov ax, values+4

push ax

mov ax, values+6

push ax

; Pop values and divide by 2

pop bx

shr bx, 1 ; divide by 2

pop bx

shr bx, 1

pop bx

shr bx, 1

pop bx

shr bx, 1

ret

;main endp

PROC1 ENDP

```
;;;;;;;;;;TASK1
```

```
; .data
```

```
; .CODE
```

```
PROC2 PROC
```

```
    mov si, offset name_str
```

```
    mov di, si
```

```
    add di, name_len-1    ; point to last character
```

```
    mov cx, name_len/2    ; loop counter for swapping
```

```
reverse_loop:
```

```
    mov al, [si]          ; get char from start
```

```
    mov bl, [di]          ; get char from end
```

```
    mov [si], bl          ; swap characters
```

```
    mov [di], al
```

```
    inc si
```

```
    dec di
```

```
    loop reverse_loop
```

```
ret
```

```
PROC2 ENDP
```

```
;;;;;;;;;;TASK2
```

```
;.DATA
```

```
;.CODE
```

```
PROC3 PROC
```

```
mov ax, val1
```

```
push ax
```

```
mov ax, val2
```

```
push ax
```

```
; Pop into opposite variables
```

```
pop ax
```

```
mov val1, ax
```

```
pop ax
```

```
mov val2, ax
```

```
ret
```

```
PROC3 ENDP
```

```
;;;;;;;;;;;;;TASK3
```

```
END MAIN
```

